# Tapuwanashe Emmanuel (Tapiwa) Magumise

Cloud Engineer | AWS, Terraform, AI

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## SUMMARY

As a cloud engineer, I spent almost three years getting 9 Zimbabwean hospitals off paper records and onto AWS. I now run PayCanary, my own SaaS product for Stripe merchants, and my nine AWS projects are public on GitHub with the code and measured results. I hold the AWS Solutions Architect and Developer Associate certifications, and an advanced diploma in modern application development, and I'm on the Codewrkx AWS Graduate Cloud Skills Accelerator.

## EXPERIENCE

**Founder & Solutions Architect** | PayCanary *Mar 2026 – Present*

*Watches Stripe accounts for risk and helps merchants deal with disputes and frozen payouts · paycanary.io*

- Built the whole stack: Next.js and TypeScript on Vercel, Supabase Postgres with row-level security, Upstash Redis and 25 scheduled jobs.
- The risk engine scores each account, groups related disputes and recommends whether each dispute is worth fighting.
- Stripe webhooks feed in account and dispute data. Every signature is checked before anything is stored.
- Found that 6 scheduled jobs had never done their work in production: they ran without database privileges, saw zero rows and reported success. Fixed them, and stopped silent write failures across 26 routes.
- Uses an LLM to draft case documents, under a spend cap and with prompt injection defences.
- Runs at 99.9% uptime, monitored with Sentry and PostHog. Tests check one customer can't read another's data, across 140 test files in Vitest and Playwright.

**Cloud Engineer** | Paperless Integrated Solutions, Bulawayo, Zimbabwe *Jun 2023 – Mar 2026*

- Moved 9 hospitals across Zimbabwe from paper folders to a digital records system, digitising 40,000 to 60,000 patient records. Some had to be scanned; others were digital files that needed cleaning up first.
- Each hospital ran its own mix of Windows versions. We moved them to AWS (EC2, S3, IAM).
- Consolidated 40,000 to 60,000 records from 9 hospital sources into one MySQL database searchable by national ID, using AWS Glue jobs that merged, de-duplicated and validated them.
- Kept 9 servers patched, backed up, monitored and protected at 99.5% availability, provisioned with Terraform and Docker.
- Designed the AWS architecture to be cost-conscious from the start, so the whole environment ran on AWS promotional credits, including the $1,000 new-account credit, and never ran out of budget in my time there. Also planned licences and hardware refreshes against budget.
- Patient data falls under Zimbabwe's Cyber and Data Protection Act, so access to records was role-based and logged.
- Trained 90 nursing staff in 18 workshops across the hospitals to use the system in their daily work.
- Before, a patient's history meant a search through folders. Afterwards nurses and doctors found it by national ID or file number, lab results included. New patients were entered on standard forms with checks such as allergy conflicts, and wards could see bed availability.

## PROJECTS

*github.com/tapiwa-magumise, each with an architecture diagram, design decisions and test results*

### Foundations and security

**Multi-Account Landing Zone** — AWS organisation in Terraform with service control policies, an organisation-wide CloudTrail under Object Lock, and sign-in through IAM Identity Center.

**Hybrid Migration and DR** — Moved an office app, its PostgreSQL database and files to AWS with 94 seconds of frozen writes and every row checked by hash. In the DR drill the app was back up in Ireland 18 min 27 s after London went down.

**Security and Cost Review Agent** — Python agent that reviews Terraform code or a live AWS account against 24 Well-Architected rules, through a read-only role.

### Data

**Serverless Sales Pipeline** — Branch sales files land in S3 and Lambda checks every row. Rejected rows are kept with the reason. SQS, DynamoDB and Athena, deployed by GitHub Actions with no stored keys.

**Edge Analytics** — Turns CloudFront logs into CDN cost by region, page speed, A/B results and bot traffic, using Step Functions, Glue and Athena.

**Online Retail Analytics** — 1,067,371 transactions analysed in DuckDB and Python, with a Power BI report for the owners.

### Applications

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| **Invoicing and Payments API** | FastAPI, PostgreSQL and Redis. A retried payment never charges twice. Load test: 10,734 requests, 0 errors, p95 of 129 ms. |
| **Event-Driven Order System** | NestJS and Spring Boot services talking over Kafka on Kubernetes. When 40 orders raced for 10 items, exactly 10 went through. |

### AI

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| **Governed RAG Assistant** | Answers UK procurement questions from 260 documents using Bedrock in London. Cites its sources and masks personal data, with a governance pack mapped to ISO/IEC 42001. Evaluation still to run. |

## SKILLS

**AWS:** EC2, S3, IAM, VPC, Lambda, Glue, Athena, RDS, DynamoDB, SQS, SNS, KMS, CloudFront, CloudWatch, Step Functions, Bedrock, Organizations

**Infrastructure as Code and CI/CD:** Terraform, GitHub Actions with OIDC, Docker, Kubernetes, Helm

**Observability:** CloudWatch, OpenTelemetry, Prometheus, Grafana, Jaeger

**Networking and OS:** VPC, Linux, Bash

**Messaging and app servers:** Kafka, Apache Tomcat

**Tools:** curl, Google Workspace

**Data:** ETL, data migration and cleansing, MySQL schema design, PostgreSQL, DuckDB

**Code:** Python, Java, TypeScript, SQL, Next.js, FastAPI

## CERTIFICATIONS & TRAINING

- AWS Certified Solutions Architect – Associate
- AWS Certified Developer – Associate
- Advanced Diploma in Modern Application Development, Alison (CPD certified), 2024
- Diploma in Learning and Development (L&D), Alison (CPD certified), 2024
- Anthropic courses: building LLM applications with the API, Model Context Protocol

*References available on request.*